Shashi Bhushan Vijay

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SUMMARY

M.Tech Computer Science Engineering student and patent holder specializing in ML pipelines, distributed systems, and backend engineering. Proven track record of shipping production-grade systems - including a patented indoor positioning solution with $R^2 = 0.978$ accuracy. Experienced across the full stack, from edge hardware integration to LLM-powered SaaS platforms.

EXPERIENCE & PROJECTS

proofStack — Agentic AI Resume Fit & Competency Verification Platform *July 2026*
Project *Repo | Live Demo*

- Architected an asynchronous, multi-stage AI engine integrating Google Gemini Pro API via FastAPI, orchestrating a **7-stage concurrent extraction pipeline** with strict latency optimizations (async I/O, prompt tuning) to reduce overall processing time by **40%** and achieve a **<3s median response time**.
- Engineered a resilient background processing pipeline to decouple heavy PDF parsing and LLM inference; executed comprehensive database optimizations (indexing, connection pooling) and an aggressive **Redis caching strategy**, accelerating query execution by **60%** and reducing database server load by **85%**.
- Developed an interactive **AI Interrogation system** using RAG (Retrieval-Augmented Generation) to conduct dynamic technical interviews on weak resume claims, automatically synthesizing verified user responses into quantified **STAR** (Situation, Task, Action, Result) bullet points.
- Designed a hybrid, dual-tier **authentication & authorization** system integrating Firebase Auth with PostgreSQL Row-Level Security (RLS) and custom short-lived anonymous JWT tokens, enabling frictionless guest evaluations while enforcing strict **multi-tenant data isolation**.
- Integrated enterprise monetization via Cashfree payment gateway and automated webhook processing, implementing custom cryptographic **HMAC-SHA256 signature verification** on raw request bodies to securely handle real-time subscription lifecycle events and enforce daily API limits.
- Built a responsive SSR frontend using **Next.js 16** (App Router), TypeScript, and Tailwind CSS; maximized loading performance via React Server Components and dynamic imports, improving **First Contentful Paint (FCP) by 45%**, with live status polling for interactive evidence matrices.

AutoML Assistant *Jan 2026*
Project *Repo | Live Demo*

- Architected an end-to-end AutoML platform using Streamlit + Python to automate data upload, preprocessing, analysis, model training, and optimization workflows.
- Designed a modular ML backend supporting **regression, classification, and time-series forecasting** with algorithms like XGBoost, Random Forest, ARIMA/SARIMAX, and optional LSTM.
- Integrated LLM-powered model recommendation (RAG) using LangChain, Groq (Llama 3.1), and FAISS to suggest suitable models based on dataset statistics and ML best practices.
- Implemented **SHAP-based explainability** (feature importance, beeswarm, waterfall, dependence plots) to make model predictions interpretable for end users.
- Exposed trained models through a **FastAPI** inference API (`/predict`, `/predict/csv`) with **persistent workspace/model management** for reusable deployment-ready ML pipelines.

Patent: Coordinated Indoor Position Determination via Multi-Stage Signal Processing *Dec 2024–Dec 2025*
VIT, Vellore, IN *Application No. 202541115892*

- Delivered an indoor localization system using Wi-Fi RSSI fingerprinting and Bluetooth integration, achieving approx. **1.6 m positioning accuracy (MAE: 1.468 m, R^2: 0.978)** in complex environments.
- Deployed an ESP32-based autonomous Wi-Fi sniffer on a robotic car, automating radio-map generation and capturing **4,277 RSSI records across 209 access points** - eliminating manual site surveys.
- Designed and optimized a Dynamic Stacking ensemble (KNN, Random Forest, XGBoost + Gradient Boosting meta-learner), cutting localization error by **approx. 43% (RMSE: 3.124 to 1.787)**.
- Engineered a real-time data pipeline using MQTT, Raspberry Pi, and Python, transforming raw Wi-Fi scans into

machine-learning-ready datasets for live location predictions.

- Implemented an **A*-based** indoor navigation module using NetworkX for optimal route generation and dynamic path recalculation from predicted user locations to destinations.

OncoFollow - LLM-Powered Clinical Workflow Platform *Apr 2026 – Jun 2026*

Project *Repo | Live Demo*

- Built an **agentic workflow** platform using Groq Cloud LLMs with role-specific **prompt engineering**, where distinct AI agents serve patients, clinicians, and admins through separate structured inference pipelines - demonstrating **multi-agent** system design.
- Designed a **serverless API layer** implementing **LLM routing** logic with role-specific prompt controls, server-side secret management, and structured JSON outputs for downstream processing.
- Engineered a rule-based triage engine across **20 symptom categories** and 4 scoring dimensions (severity, duration, frequency, trend); incorporated red-flag detection and override logic, achieving **99% sensitivity** for emergency-case interception.
- Anchored all AI outputs in NCCN/ASCO clinical guidelines with citation-backed **document summarization** responses; implemented output validation and blocked-phrase filtering for HIPAA-adjacent compliance.
- Crafted real-time **analytics dashboards** (Recharts) with AI-powered trend analysis and NEWS2 acuity scoring for actionable **decision support**.

EDUCATION

Vellore Institute of Technology, Vellore *Expected April 2027*

Integrated M.Tech in Computer Science Engineering GPA: 7.0/10

Relevant Coursework: Machine Learning, Distributed Systems, Data Structures & Algorithms, Database Management, Computer Networks, Operating Systems, Cloud Computing

SM Public School, Haridwar *2020–21*

Junior College (Class XII) 91.2%

SM Public School, Haridwar *2018–19*

High School (Class X) 87%

TECHNICAL SKILLS

Languages & Core: Python, C++, Java, JavaScript, SQL

Backend & Systems: FastAPI, REST APIs, Microservices, MQTT, Docker, CI/CD

Databases: PostgreSQL, SQLite, Redis | **Tools & Cloud:** Git, AWS, Streamlit, Power BI

ML & AI: Scikit-learn, XGBoost, TensorFlow, PyTorch, LSTM, SHAP, RAG, FAISS, LangChain, Groq

LEADERSHIP & EXTRACURRICULARS

Sponsorship Coordinator, ISTE Chapter – VIT Vellore

- Secured **Rs. 1,00,000+ in corporate sponsorships** for flagship hackathon (Horizon) by pitching to **100+ founders and startups**; orchestrated end-to-end event logistics and developed outreach frameworks that established strategic industry connections for future initiatives.

ACHIEVEMENTS

1st Place, ARC Hackathon (ARC Lab, VIT) – Created a GUI-based AutoML tool that automatically benchmarks and recommends the best ML model for any given dataset, winning against competing teams.

CERTIFICATIONS

Stock Market Basics & Technical Analysis – View Certificate

Designing Seamless Notification Service – View Certificate